

Norway

SLEIPNER

Volve F

F-11

15/9-F-11

Design: 15/9-F-11

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-11		
Well Position	+N/-S	4.65 m	Northing:
	+E/-W	-0.93 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.9566 N
			Longitude:
			1° 53' 14.8667 E
			Water Depth:
			91.00 m

Wellbore	15/9-F-11		
Magnetics	Model Name	Sample Date	Declination
			(°)
	MAGNETICREFERENCE	02.12.2010	-1.68
			Dip Angle
			(°)
			71.63
			Field Strength
			(nT)
			50,437

Design	15/9-F-11		
Audit Notes:			
Version:	1.0	Phase:	ACTUAL
		Tie On Depth:	145.90
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	4.65	-0.93
			Direction
			(°)
			62.39

Survey Program	Date	01.07.2013			
From	To	Survey (Wellbore)	Tool Name	Description	
(m)	(m)				
150.00	230.00	F-11 T2 SDI Keeper Gyro (15/9-F-11 T2)	Keeper, cont	Gyro Tool from SDC	
238.50	305.50	F-11 Gyro SDC 202-306m after reaming (1	Gyro, old	Other Gyro Tools	
319.20	319.20	F-11 GyroTrak 26" (15/9-F-11)	NOT_useMWD gyro, Baker & Baker Hughes' MWD gyro service (GyroTrak)		

Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
145.90	0.00	0.00	145.90	4.65	-0.93	0.00	0.000	0.000	0.000	
150.00	0.09	227.76	150.00	4.65	-0.93	0.00	0.659	0.659	0.000	
160.00	0.20	225.57	160.00	4.63	-0.95	-0.03	0.330	0.330	-6.570	
170.00	0.19	227.59	170.00	4.61	-0.97	-0.06	0.036	-0.030	6.060	
180.00	0.23	231.32	180.00	4.58	-1.00	-0.10	0.127	0.120	11.190	
190.00	0.23	233.02	190.00	4.56	-1.03	-0.14	0.020	0.000	5.100	
200.00	0.22	235.12	200.00	4.54	-1.06	-0.17	0.039	-0.030	6.300	
210.00	0.24	234.55	210.00	4.51	-1.10	-0.21	0.060	0.060	-1.710	
220.00	0.34	239.76	220.00	4.49	-1.14	-0.26	0.310	0.300	15.630	

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Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
230.00	0.42	244.41	230.00	4.46	-1.20	-0.33	0.257	0.240	13.950	
238.50	0.49	217.49	238.50	4.41	-1.25	-0.39	0.785	0.247	-95.012	
251.30	0.57	204.82	251.30	4.31	-1.31	-0.49	0.331	0.187	-29.695	
261.30	0.60	167.13	261.30	4.22	-1.32	-0.55	1.137	0.090	-113.070	
263.50	0.66	165.04	263.50	4.19	-1.31	-0.55	0.876	0.818	-28.500	
275.30	0.56	208.46	275.30	4.08	-1.32	-0.62	1.171	-0.254	110.390	
278.30	0.56	198.61	278.30	4.05	-1.33	-0.64	0.962	0.000	-98.500	
288.30	0.49	195.01	288.30	3.96	-1.36	-0.70	0.232	-0.210	-10.800	
291.70	0.57	184.79	291.70	3.93	-1.36	-0.72	1.090	0.706	-90.176	
301.70	0.58	167.02	301.70	3.83	-1.36	-0.76	0.534	0.030	-53.310	
305.50	0.68	161.94	305.50	3.79	-1.35	-0.77	0.904	0.789	-40.105	
319.20	0.70	195.01	319.19	3.63	-1.34	-0.84	0.861	0.044	72.416	
347.00	0.70	195.01	346.99	3.31	-1.43	-1.07	0.000	0.000	0.000	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name		Casing Diameter (in)	Hole Diameter (in)
202.00	202.00	30"		30.000	36.000

Checked By: _____ Approved By: _____ Date: _____